

# Data Analysis for Heart disease Patients

This report consists of Factors that cause Heart-Diseases like Smoking, Cholesterol, Diabetes, BMI and Exercise level, Smoker with Heart Disease, it also consists of the total how many patients are suffering from Heart-diseases by comparing their blood pressure and also so many factors like Gender, Age group...

**Project Title: Heart Disease Risk Analysis**

**Role: Data Analyst (Excel-Based Analysis)**

## 1. Business Objective

The purpose of this analysis is to identify the major factors contributing to heart disease among patients and support hospital decision-making for risk reduction and preventive strategies.

The hospital is particularly interested in:

- Understanding high-risk groups
- Evaluating impact of lifestyle factors (like smoking)
- Supporting preventive healthcare planning

## Data Overview

The dataset contains patient medical and lifestyle information related to heart disease.

| Category              | Data Fields                                  |
|-----------------------|----------------------------------------------|
| Demographics          | Age, Gender                                  |
| Lifestyle             | Smoking, Exercise level                      |
| Clinical Measures     | Blood Pressure, Cholesterol, BMI, Heart Rate |
| Medical Conditions    | Diabetes                                     |
| Diagnostic Indicators | ECG, Exercise pain, Vessel blockage          |
| Outcome               | Heart Disease (Yes/No)                       |

## Tools & Method Used

The analysis was performed using **Microsoft Excel** with basic analytical functions:

- Filtering & Sorting

- COUNTIF
- SUMIF
- IF logic
- Percentage calculations

## Key Findings

### High Impact Risk Factors Identified

| Factor              | Observation                           |
|---------------------|---------------------------------------|
| Smoking             | Strong association with heart disease |
| High Blood Pressure | Common in heart disease patients      |
| High Cholesterol    | Major contributor to artery blockage  |
| Diabetes            | Increases vessel damage risk          |
| Age above 50        | Higher disease occurrence             |

### Population Insights

- Heart disease cases are higher in **older patients**
- Smokers show a **greater proportion of heart disease**
- Diabetic patients have increased cardiovascular risk
- People with low physical activity show higher cases
- 

### Business Impact

The analysis shows heart disease is **multi-factorial**. Focusing on only one risk factor will give partial improvement but not major reduction.

### Recommendations

To significantly reduce heart disease:

1. Smoking cessation programs
2. Blood pressure control initiatives
3. Cholesterol monitoring

4. Diabetes screening
5. Lifestyle awareness (diet + exercise)

A combined approach is required.

### **Limitations**

- Analysis done using basic Excel functions
- No predictive modeling used
- Results are pattern-based, not medically diagnostic

## QUESTIONS:

### Descriptive Analysis:

#### **1.How many Females were Smoking?**

Ans. Total Females = 240

Total Females who Smoke-102

#### **2.At the Age of 45 below how many Females have heart diseases?**

Ans. 11 females have cardiac disease.

#### **3. Check whether they are diabetic or not?**

Ans. Diabetic – 151 and Non-diabetic -349

#### **4.If Females are smoking how many are suffer from cardiac disease?**

Ans. 48 Females who smokes and suffers from cardiac disease.

#### **5.Average BMI of both Male and Female.**

Ans. Average BMI – 29.216

Max BMI -40

MIN BMI -18

#### **6.Total How many have higher heart rate rather than normal Heart rate?**

Ans. Compare to normal Heart rate -72, 356 members has higher Heartrate.

#### **7.How many people are of age above 60?**

Ans. Total 208 people are of age above 60.

**8.What is the percentage of people with Heart-Disease? Both in Female and Male also the total smokers in male and female?**

Ans. Total females who having heart disease = 68

Total males who smoke and having heart disease = 72

Percentage of smokers who having heart disease = 73.57143%

**9.Total number of Diabetic with heart disease and without heart disease?**

Ans. Total no of people with heart disease – 73

Total no of people with dose not have heart disease -67

**10.Total Smokers who faces with Heart-disease and Non-smokers with heart disease?**

Ans. Non-smokers who have heart disease = 37

Smokers who have heart disease = 67.

**11.What is average age of patients who doesn't has heart-disease, who has the heart disease?**

Ans. Average age of people who doesn't have heart disease – 53

. Average age of people who does have heart disease- 59

**12.What is the Average Cholesterol of total patients, Max and Min value?**

Ans. Average cholesterol value – 224.156

Max value-300

Min value -150

**13.How many people were at age above 50?**

Ans. Total 317 people were at age above 50.

**14.How many total smokers who has high risk and non-smokers who don't have high risk?**

Ans. smokers who have high risk -212 and non-smokers who don't have high risk-288

**15.Above 240 cholesterol how many has heart diseases?**

Ans. 200members

**16.How many people who workout less than 2 hrs and who workout more than 2 hrs has heart disease?**

Ans. people who have workout less than 2 hrs – 57 members and who have workout more than 2 hrs- 83 members have heart disease.

## **Predictive:**

**1.If 65 years old patient of 260 cholesterol and 150 BP also smokes. Does he have high risk of heart disease or not?**

Ans. A very high risk of Heart-Disease but cannot be confirmed until using the Diagnostics Test.

**2.What are common combinatory factors that causing heart disease?**

Ans. Common factors that causing heart diseases are High BP, Cholesterol, Smoking and Diabetes

**3.If the patient has normal cholesterol and very high Blood pressure at age above 35. According to this dataset does he have high risk or low risk of heart disease?**

Ans. The patient has normal cholesterol and very high Blood pressure have high risk of heart disease according to this dataset.

**4.Among the patients how many who has heart disease are smokers?**

Ans. 62 Members who were both diabetic and Smoker have heart disease.

**5.What proportion have heart disease is categorized?**

Ans. Diabetic: Non-Diabetic - 73:67

Smoker: Non-Smoker-103:37

Female: Male-68:72

**6.Which one section have high risk of heart disease?**

Ans. Smokers shows the high risk of Heart-disease.

**7.Which groups should the hospital target? which one should prefer more medical assistance for preventive purpose?**

Ans. Among smokers, diabetes, Blood Pressure, High heartrate the hospitals target for future preventions

Smokers>Diabetic>High BP>Heart rate.

**8.If the hospital works only awareness program, which should be focused on smoking does it help on reducing the rate of heart disease more?**

Ans. If we focusing on smoking it can reduce the heart diseases rate. Along side the hospital should consider diabetes and High Bp patients.

**9.Which group should receive free heart screening tests compared to others?**

Ans. Free Treatment should be provided for smokers with diabetes and high BP.

**10.Do doctors emphasize cholesterol and Blood pressure does it reduce heart disease completely?**

Ans. Proper medical which can reduce the high risk of heart disease.

**11. If the resources are limited which two factors affects more on heart disease so that the rate of heart disease can be at control?**

Ans. If the resources are limited then targeting smoking and diabetes will reduce the risk rate of heart disease.

## Prescriptive:

**1.If 45 years old Man who is a nonsmoker has BMI of 28 and no diabetes with good exercise level. According to dataset does he have risk of heart disease or not?**

Ans. A 45year-old man smoking with no Diabetes and good exercise habits falls into low-risk group according to dataset and impossible chance of having heart disease.

**2.If the two patients p1 and p2**

| Name        | P1           | P2         |
|-------------|--------------|------------|
| Age         | 70           | 55         |
| Smoking     | smoker       | Non smoker |
| Cholesterol | 250          | 210        |
| Diabetes    | Non diabetic | Diabetic   |

**Which patient has statically Resemble the heart disease group more?**

Ans. Among two patients p1 and p2. Both has equal chances of risk of heart disease because p1 has gotten old and p2 is non smoker but he is at age of 55 the cholesterol level is insane.

**3. Let the hospital goal is to reduce the percentage of heart disease to 10, as if they convert all the smokers into non-smokers. Does percentage reaches to 10 or not?**

Ans. No I don't think so if they all converted to Nonsmoker's they were still the other factors which can cause heart disease like High BP, Cholesterol, Diabetes and Age.

**4. There is a corporate Company who conducted annual checkup, From data analysis what early precautions so can risk of heart disease can be reduced?**

Ans. Most of Corporate Companies employees have high risk of heart disease from workload and stress I think some of might have addiction for smoking to have some relief from work pressure, some might have High Bp and Diabetes. Most of them are smokers. Awareness programs about smoking should be conducted and also stress-free activities like dancing, some sort of programs based of workout activities should be conducted.

**5. If the patient improves his workout 0 hrs to 5 hrs per week. But the remaining values are same on the dataset. Suggest the risk category based on the data set?**

Ans. If the patients improve from workouts 0hrs to 5hrs also if doesn't leave smoking or eating junk foods which cause obesity. Still it will be a still high risk of heart disease.

From further Analysis we can see Percentage of smokers who having heart disease - 73.57143

Total cholesterol who having diabetes = 32609

Total cholesterol who doesn't having diabetes = 79469

Total sum of BMI who have heart disease- 4502.7

## **Formulas that I used for calculations:**

1. =COUNTIF(range, condition)

2. =IF(condition, value\_if\_true, value\_if\_false)

3. =SUMIF(range, condition, sum\_range)

4. =IF(AND(condition, value\_if\_true, value\_if\_false))

